

# INSTALLATION INSTRUCTIONS

## AIR-COOLED CONDENSING UNITS

(-)ANL-\*AZ MODEL SERIES – 13 SEER

(-)APM-JAZ MODEL SERIES – 14.5 SEER



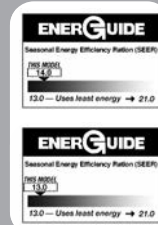
FEATURING INDUSTRY STANDARD  
R-410A REFRIGERANT **R-410A**



RECOGNIZE THIS SYMBOL AS AN INDICATION OF IMPORTANT SAFETY INFORMATION!

### ⚠ WARNING

THESE INSTRUCTIONS ARE INTENDED AS AN AID TO QUALIFIED, LICENSED SERVICE PERSONNEL FOR PROPER INSTALLATION, ADJUSTMENT AND OPERATION OF THIS UNIT. READ THESE INSTRUCTIONS THOROUGHLY BEFORE ATTEMPTING INSTALLATION OR OPERATION. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN IMPROPER INSTALLATION, ADJUSTMENT, SERVICE OR MAINTENANCE POSSIBLY RESULTING IN FIRE, ELECTRICAL SHOCK, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



Accredited by the RvA



ISO 9001:2008

Certificate Number: 30164

**DO NOT DESTROY THIS MANUAL**

**PLEASE READ CAREFULLY AND KEEP IN A SAFE PLACE FOR FUTURE REFERENCE BY A SERVICEMAN**



[ ] INDICATES METRIC CONVERSIONS

92-21354-51-13  
SUPERSEDES 92-21354-51-12

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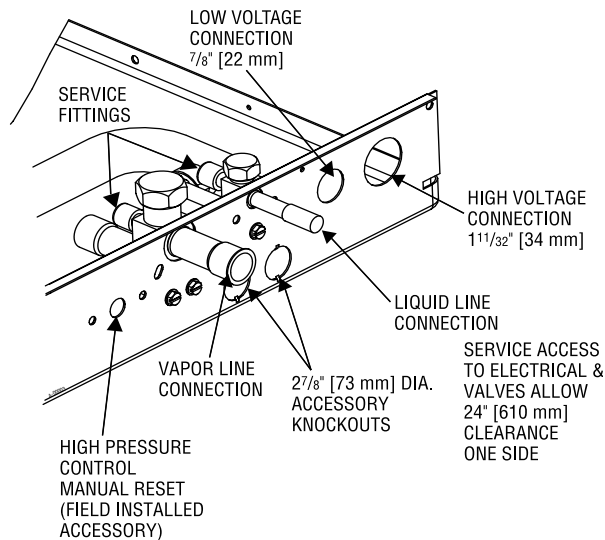
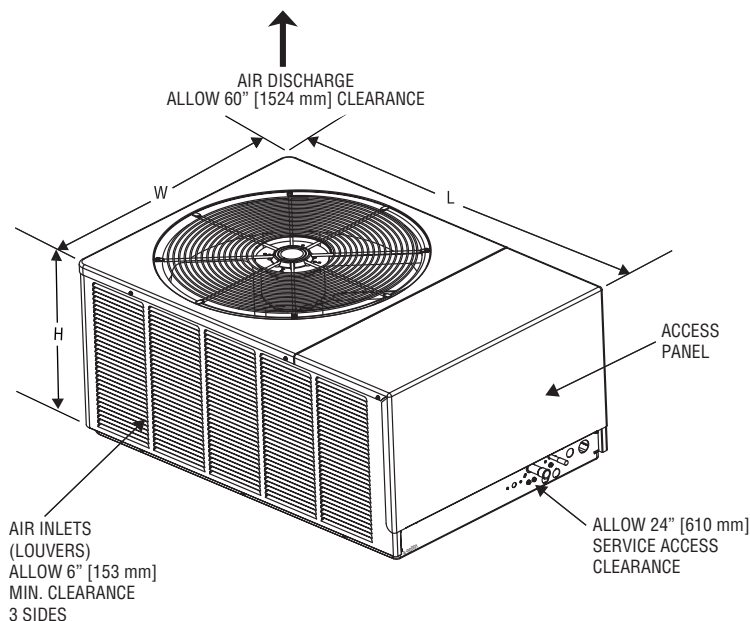
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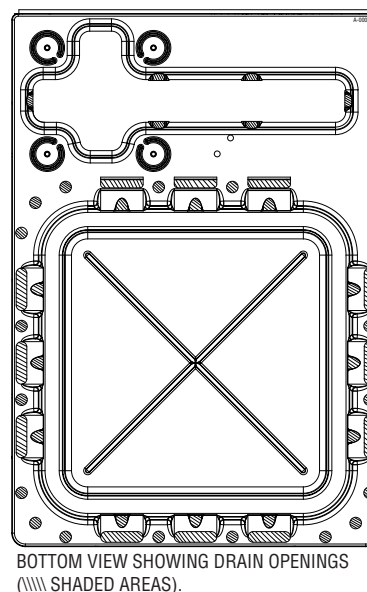
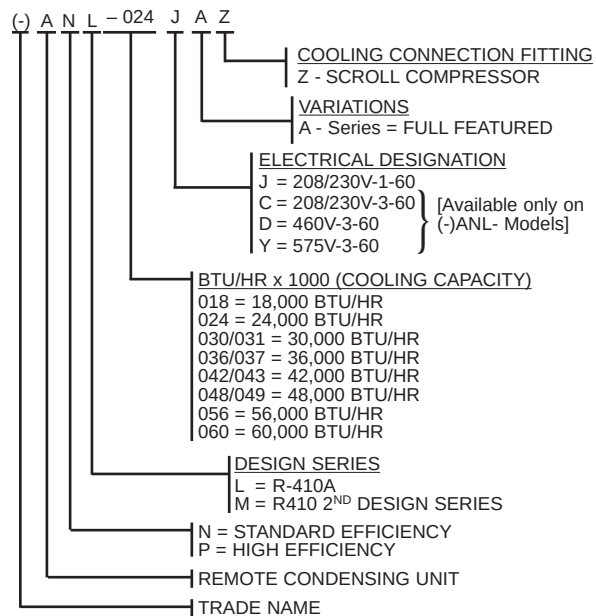
## CHECKING PRODUCT RECEIVED

Upon receiving unit, inspect it for any shipping damage. Claims for damage, either apparent or concealed, should be filed immediately with the shipping company. Check condensing unit model number, electrical characteristics and accessories to determine if they are correct. Check system components (evaporator coil, condensing unit, evaporator blower, etc.) to make sure they are properly matched.

**FIGURE 1**  
DIMENSIONS AND INSTALLATION CLEARANCES



## UNIT MODEL NUMBER EXPLANATION



## DIMENSIONAL DATA

| CONDENSING UNIT MODEL (-)ANL | 018  | 024/031 | 043/037 | 049  | 060  |
|------------------------------|------|---------|---------|------|------|
| LENGTH "H" (INCHES)          | 19"  | 19"     | 23"     | 29"  | 33"  |
| LENGTH "L" (INCHES)          | 35½" | 40½"    | 44⅜"    | 44⅜" | 44⅜" |
| WIDTH "W" (INCHES)           | 24¾" | 27⅝"    | 31½"    | 31½" | 31½" |

| CONDENSING UNIT MODEL (-)APM | 018JAZ | 024JAZ<br>030JAZ | 036JAZ/042JAZ<br>048JAZ/056JAZ/060JAZ |
|------------------------------|--------|------------------|---------------------------------------|
| LENGTH "H" (INCHES)          | 19"    | 29"              | 33"                                   |
| LENGTH "L" (INCHES)          | 40½"   | 44⅜"             | 44⅜"                                  |
| WIDTH "W" (INCHES)           | 27⅝"   | 31½"             | 31½"                                  |

**TABLE 1**  
**(-)ANL ELECTRICAL AND PHYSICAL DATA**

| Model Number<br>(-)ANL- | ELECTRICAL                                 |                                   |                                     |                                            |                                           |                                 |                    | PHYSICAL                  |             |              |                                      |                  |                       |
|-------------------------|--------------------------------------------|-----------------------------------|-------------------------------------|--------------------------------------------|-------------------------------------------|---------------------------------|--------------------|---------------------------|-------------|--------------|--------------------------------------|------------------|-----------------------|
|                         | Phase<br>Frequency (Hz)<br>Voltage (Volts) | Compressor                        |                                     | Fan Motor<br>Full Load<br>Amperes<br>(FLA) | Minimum<br>Circuit<br>Ampacity<br>Amperes | Fuse or HACR<br>Circuit Breaker |                    | Outdoor Coil              |             |              | Refrig.<br>Per<br>Circuit<br>Oz. [g] | Weight           |                       |
|                         |                                            | Rated<br>Load<br>Amperes<br>(RLA) | Locked<br>Rotor<br>Amperes<br>(LRA) |                                            |                                           | Minimum<br>Amperes              | Maximum<br>Amperes | Face Area<br>Sq. Ft. [m²] | No.<br>Rows | CFM<br>[L/s] |                                      | Net<br>Lbs. [kg] | Shipping<br>Lbs. [kg] |
| 018JAZ                  | 1-60-208/230                               | 9/9                               | 48                                  | 0.6                                        | 12/12                                     | 15/15                           | 20/20              | 9.07 [0.84]               | 1           | 1775 [838]   | 70 [1984]                            | 130 [59]         | 140 [63.5]            |
| 024JAZ                  | 1-60-208/230                               | 12.8/12.8                         | 58.3                                | 0.6                                        | 17/17                                     | 20/20                           | 25/25              | 11 [1.02]                 | 1           | 1920 [906]   | 72 [2041]                            | 140 [63.5]       | 150 [68]              |
| 031JAZ                  | 1-60-208/230                               | 14.1/14.1                         | 73                                  | 0.6                                        | 19/19                                     | 25/25                           | 30/30              | 11.1 [1.02]               | 1           | 1920 [906]   | 83 [2353]                            | 142 [54]         | 152 [70]              |
| 037CAZ                  | 3-60-208-230                               | 13.2/13.2                         | 88                                  | 1.2                                        | 18/18                                     | 25/25                           | 30/30              | 16.1 [1.5]                | 1           | 2300 [1085]  | 106 [3005]                           | 160 [72]         | 170 [77]              |
| 037DAZ                  | 3-60-460                                   | 6.0                               | 44                                  | 0.6                                        | 9                                         | 15                              | 15                 | 16.1 [1.5]                | 1           | 2300 [1085]  | 106 [3005]                           | 160 [72]         | 170 [77]              |
| 037JAZ                  | 1-60-208/230                               | 17.9/17.9                         | 112                                 | 1.2                                        | 24/24                                     | 30/30                           | 40/40              | 16.1 [1.5]                | 1           | 2300 [1085]  | 106 [3005]                           | 160 [72]         | 170 [77]              |
| 043CAZ                  | 3-60-208-230                               | 13.5/13.5                         | 83.1                                | 1.2                                        | 19/19                                     | 25/25                           | 30/30              | 17.26 [1.6]               | 1           | 3200 [1510]  | 115 [3260]                           | 205 [93]         | 215 [97]              |
| 043DAZ                  | 3-60-460                                   | 6.0                               | 44                                  | 0.6                                        | 9                                         | 15                              | 15                 | 17.26 [1.6]               | 1           | 3200 [1510]  | 115 [3260]                           | 205 [93]         | 215 [97]              |
| 043JAZ                  | 1-60-208/230                               | 19.9/19.9                         | 109                                 | 1.2                                        | 27/27                                     | 35/35                           | 45/45              | 17.26 [1.6]               | 1           | 3200 [1510]  | 115 [3260]                           | 205 [93]         | 215 [97]              |
| 049CAZ                  | 3-60-208-230                               | 13.7/13.7                         | 83.1                                | 1.2                                        | 19/19                                     | 25/25                           | 30/30              | 20.1 [1.8]                | 1           | 3200 [1510]  | 132 [3742]                           | 235 [106]        | 245 [111]             |
| 049DAZ                  | 3-60-460                                   | 6.2                               | 41                                  | 0.6                                        | 9                                         | 15                              | 15                 | 20.1 [1.8]                | 1           | 3200 [1510]  | 132 [3742]                           | 235 [106]        | 245 [111]             |
| 049JAZ                  | 1-60-208/230                               | 21.8/21.8                         | 117                                 | 1.2                                        | 29/29                                     | 35/35                           | 50/50              | 20.1 [1.8]                | 1           | 3200 [1510]  | 132 [3742]                           | 235 [106]        | 245 [111]             |
| 049YAZ                  | 3-60-575                                   | 4.8                               | 33                                  | 0.5                                        | 7                                         | 15                              | 15                 | 20.1 [1.8]                | 1           | 3200 [1510]  | 132 [3742]                           | 235 [106]        | 245 [111]             |
| 060CAZ                  | 3-60-208-230                               | 15.6/15.6                         | 110                                 | 1.2                                        | 21/21                                     | 25/25                           | 35/35              | 23.01 [2.14]              | 1           | 3500 [1652]  | 180 [5103]                           | 250 [113.4]      | 260 [117.9]           |
| 060DAZ                  | 3-60-460                                   | 7.8                               | 52                                  | 0.6                                        | 11                                        | 15                              | 15                 | 23.01 [2.14]              | 1           | 3500 [1652]  | 180 [5103]                           | 250 [113.4]      | 260 [117.9]           |
| 060JAZ                  | 1-60-208/230                               | 26.3/26.3                         | 134                                 | 1.2                                        | 35/35                                     | 45/45                           | 60/60              | 23.01 [2.14]              | 1           | 3500 [1652]  | 180 [5103]                           | 250 [113.4]      | 260 [117.9]           |
| 060YAZ                  | 3-60-575                                   | 5.8                               | 38.9                                | 0.5                                        | 8                                         | 15                              | 15                 | 23.01 [2.14]              | 1           | 3500 [1652]  | 180 [5103]                           | 250 [113.4]      | 260 [117.9]           |

**TABLE 2**  
(-)APM ELECTRICAL AND PHYSICAL DATA

| Model Number<br>RAPM- | ELECTRICAL                                 |                                |                                  |                                            |                                           |                                 |                    | PHYSICAL                  |             |              |                                      |                  |                       |
|-----------------------|--------------------------------------------|--------------------------------|----------------------------------|--------------------------------------------|-------------------------------------------|---------------------------------|--------------------|---------------------------|-------------|--------------|--------------------------------------|------------------|-----------------------|
|                       | Phase<br>Frequency (Hz)<br>Voltage (Volts) | Compressor                     |                                  | Fan Motor<br>Full Load<br>Amperes<br>(FLA) | Minimum<br>Circuit<br>Ampacity<br>Amperes | Fuse or HACR<br>Circuit Breaker |                    | Outdoor Coil              |             |              | Refrig.<br>Per<br>Circuit<br>Oz. [g] | Weight           |                       |
|                       |                                            | Rated Load<br>Amperes<br>(RLA) | Locked Rotor<br>Amperes<br>(LRA) |                                            |                                           | Minimum<br>Amperes              | Maximum<br>Amperes | Face Area<br>Sq. Ft. [m²] | No.<br>Rows | CFM<br>[L/s] |                                      | Net<br>Lbs. [kg] | Shipping<br>Lbs. [kg] |
|                       |                                            |                                |                                  |                                            |                                           |                                 |                    |                           |             |              |                                      |                  |                       |
| Rev. 3/11/2010        |                                            |                                |                                  |                                            |                                           |                                 |                    |                           |             |              |                                      |                  |                       |
| 018JAZ                | 1-60-208/230                               | 9/9                            | 48                               | 1.9                                        | 14/14                                     | 20/20                           | 20/20              | 11 [1.02]                 | 1           | 1955 [923]   | 82 [2325]                            | 138.5 [62.8]     | 160.5 [72.8]          |
| 024JAZ                | 1-60-208/230                               | 13.5/13.5                      | 58.3                             | 1.1                                        | 18/18                                     | 25/25                           | 30/30              | 20 [1.86]                 | 1           | 3295 [1555]  | 128 [3629]                           | 192.5 [87.3]     | 214.5 [97.3]          |
| 030JAZ                | 1-60-208/230                               | 12.8/12.8                      | 64                               | 1.9                                        | 18/18                                     | 25/25                           | 30/30              | 20 [1.86]                 | 1           | 3645 [1720]  | 129 [3657]                           | 193 [87.5]       | 214 [97.1]            |
| 036JAZ                | 1-60-208/230                               | 16/16                          | 79                               | 1.9                                        | 22/22                                     | 30/30                           | 35/35              | 23.01 [2.14]              | 1           | 3680 [1737]  | 146 [4139]                           | 197.5 [89.6]     | 227 [103]             |
| 042JAZ                | 1-60-208/230                               | 17.9/17.9                      | 112                              | 2.8                                        | 26/26                                     | 30/30                           | 40/40              | 23.01 [2.14]              | 1           | 3925 [1852]  | 152 [4309]                           | 223 [101.2]      | 256 [116.1]           |
| 048JAZ                | 1-60-208/230                               | 21.8/21.8                      | 117                              | 2.8                                        | 31/31                                     | 40/40                           | 50/50              | 23.01 [2.14]              | 2           | 3685 [1739]  | 203 [5755]                           | 264 [119.8]      | 298 [135.2]           |
| 056JAZ                | 1-60-208/230                               | 21.4/21.4                      | 135                              | 2.8                                        | 30/30                                     | 35/35                           | 50/50              | 23 [2.14]                 | 2           | 3685 [1739]  | 255 [7229]                           | 269.5 [122.2]    | 306.5 [139]           |
| 060JAZ                | 1-60-208/230                               | 26.4/26.4                      | 134                              | 2.8                                        | 36/36                                     | 45/45                           | 60/60              | 23 [2.14]                 | 2           | 4350 [2053]  | 262 [7428]                           | 269.5 [122.2]    | 306.5 [139]           |

## WARNING

THE MANUFACTURER'S WARRANTY DOES NOT COVER ANY DAMAGE OR DEFECT TO THE AIR CONDITIONER CAUSED BY THE ATTACHMENT OR USE OF ANY COMPONENTS, ACCESSORIES OR DEVICES (OTHER THAN THOSE AUTHORIZED BY THE MANUFACTURER) INTO, ONTO OR IN CONJUNCTION WITH THE AIR CONDITIONER. YOU SHOULD BE AWARE THAT THE USE OF UNAUTHORIZED COMPONENTS, ACCESSORIES OR DEVICES MAY ADVERSELY AFFECT THE OPERATION OF THE AIR CONDITIONER AND MAY ALSO ENDANGER LIFE AND PROPERTY. THE MANUFACTURER DISCLAIMS ANY RESPONSIBILITY FOR SUCH LOSS OR INJURY RESULTING FROM THE USE OF SUCH UNAUTHORIZED COMPONENTS, ACCESSORIES OR DEVICES.

### MATCH ALL COMPONENTS:

- OUTDOOR UNIT
- INDOOR COIL/METERING DEVICE
- INDOOR AIR HANDLER/FURNACE
- REFRIGERANT LINES

## GENERAL

The information contained in this manual has been prepared to assist in the proper installation, operation and maintenance of the air conditioning system. Improper installation, or installation not made in accordance with these instructions, can result in unsatisfactory operation and/or dangerous conditions, and can cause the related warranty not to apply.

Read this manual and any instructions packaged with separate equipment required to make up the system prior to installation. Retain this manual for future reference.

To achieve optimum efficiency and capacity, the indoor cooling coils listed in the condensing unit specification sheet should be used.

## APPLICATION

Before installing any air conditioning equipment, a duct analysis of the structure and a heat gain calculation must be made. A heat gain calculation begins by measuring all external surfaces and openings that gain heat from the surrounding air and quantifying that heat gain. A heat gain calculation also calculates the extra heat load caused by sunlight and by humidity removal.

There are several factors that the installers must consider:

- Outdoor unit location
- System refrigerant charge
- Indoor unit blower speed
- System air balancing
- Proper equipment evacuation
- Indoor unit airflow
- Supply and return air duct design and sizing
- Diffuser and return air grille location and sizing

## CORROSIVE ENVIRONMENT

The metal parts of this unit may be subject to rust or deterioration if exposed to a corrosive environment. This oxidation could shorten the equipment's useful life. Corrosive elements include, but are not limited to, salt spray, fog or mist in seacoast areas, sulphur or chlorine from lawn watering systems, and various chemical contaminants from industries such as paper mills and petroleum refineries.

If the unit is to be installed in an area where contaminants are likely to be a problem, special attention should be given to the equipment location and exposure.

- Avoid having lawn sprinkler heads spray directly on the unit cabinet.
- In coastal areas, locate the unit on the side of the building away from the waterfront.
- Shielding provided by a fence or shrubs may give some protection, but cannot violate minimum airflow and service access clearances.
- Elevating the unit off its slab or base enough to allow air circulation will help avoid holding water against the basepan.

Regular maintenance will reduce the build-up of contaminants and help to protect the unit's finish.

## **WARNING**

**DISCONNECT ALL POWER TO UNIT BEFORE STARTING MAINTENANCE. FAILURE TO DO SO CAN CAUSE ELECTRICAL SHOCK RESULTING IN SEVERE PERSONAL INJURY OR DEATH.**

- Frequent washing of the cabinet, fan blade and coil with fresh water will remove most of the salt or other contaminants that build up on the unit.
- Regular cleaning and waxing of the cabinet with an automobile polish will provide some protection.
- A liquid cleaner may be used several times a year to remove matter that will not wash off with water.

Several different types of protective coatings are offered in some areas. These coatings may provide some benefit, but the effectiveness of such coating materials cannot be verified by the equipment manufacturer.

## **LOCATING UNIT**

### **CONDENSER LOCATION**

Consult local and national building codes and ordinances for special installation requirements. Following location information will provide longer life and simplified servicing of the outdoor condenser.

**NOTE:** These units must be installed outdoors. No ductwork can be attached, or other modifications made, to the discharge grille. Modifications will affect performance or operation.

### **OPERATIONAL ISSUES**

- **IMPORTANT:** Locate the condenser in a manner that will not prevent, impair or compromise the performance of other equipment horizontally installed in proximity to the unit. Maintain all required minimum distances to gas and electric meters, dryer vents, exhaust and inlet openings. In the absence of National Codes, or manufacturers' recommendations, local code recommendations and requirements will take precedence.
- Refrigerant piping and wiring should be properly sized and kept as short as possible to avoid capacity losses and increased operating costs.
- Locate the condenser where water run off will not create a problem with the equipment. Position the unit away from the drip edge of the roof whenever possible. Units are weatherized, but can be affected by the following:
  - o Water pouring into the unit from the junction of rooflines, without protective guttering. Large volumes of water entering the condenser while in operation can impact fan blade or motor life, and coil damage may occur to a heat pump if moisture cannot drain from the unit under freezing conditions.
- Closely follow clearance recommendations on Page 3.
  - o 24" to the service panel access
  - o 60" above condenser fan discharge (unit top) to prevent recirculation
  - o 6" to condenser coil grille air inlets (per condenser).

### **FOR CONDENSERS WITH SPACE LIMITATIONS**

In the event that a space limitation exists, we will permit the following clearances:

**Single Unit Applications:** Clearances below 6 inches will reduce unit capacity and efficiency. Do not reduce the 60-inch discharge, or the 24-inch service clearances.

**Multiple Unit Applications:** When multiple condenser grille sides are aligned, a 6-inch per unit clearance is recommended, for a total of 12" between two units. Two combined clearances below 12 inches will reduce capacity and efficiency. Do not reduce the 60-inch discharge, or 24-inch service, clearances.

- Do not obstruct the bottom drain opening in the condenser base pan. It is essential to provide defrost condensate drainage to prevent possible refreezing of the condensation. Provide a base pad for mounting the unit, which is slightly pitched away from the structure. Route condensate off the base pad to an area which will not become slippery and result in personal injury.



## CUSTOMER SATISFACTION ISSUES

- The condenser should be located away from the living, sleeping and recreational spaces of the owner and those spaces on adjoining property.
- To prevent noise transmission, the mounting pad for the outdoor unit should not be connected to the structure, and should be located sufficient distance above grade to prevent ground water from entering the unit.

## PROPER INSTALLATION

Proper sizing and installation of equipment is critical to achieve optimal performance. Use the information in this Installation Instruction Manual and reference the applicable Engineering Specification Sheet when installing this product.

**IMPORTANT:** This product has been designed and manufactured to meet ENERGY STAR® criteria for energy efficiency when matched with appropriate coil components. However, proper refrigerant charge and proper air flow are critical to achieve rated capacity and efficiency. Installation of this product should follow the manufacturer's refrigerant charging and air flow instructions. **Failure to confirm proper charge and airflow may reduce energy efficiency and shorten equipment life.**

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## UNIT MOUNTING

If elevating the condensing unit, either on a flat roof or on a slab, observe the following guidelines.

- The base pan provided elevates the condenser soil 3/4" above the base pad.
- If elevating a unit on a flat roof, use 4" x 4" (or equivalent) stringers positioned to distribute unit weight evenly and prevent noise and vibration.

**NOTE:** Do not block drain openings shown in Figure 1.

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## FACTORY-PREFERRED TIE-DOWN METHOD FOR OUTDOOR UNITS

**IMPORTANT:** The Manufacturer approved/recommended method is a guide to securing equipment for wind and seismic loads. Other methods might provide the same result, but the Manufacturer method is the only one endorsed by Manufacturer for securing equipment where wind or earthquake damage can occur. Additional information is available in the PTS (Product Technical Support) section of the Manufacturer website Rheemote.net and can be found as a listing under each outdoor model. If you do not have access to this site, your Distributor can offer assistance.

## REFRIGERANT CONNECTIONS

All units are factory charged with Refrigerant 410A. All models are supplied with service valves. Keep tube ends sealed until connection is to be made to prevent system contamination.

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## TOOLS REQUIRED FOR INSTALLING & SERVICING R-410A MODELS

### Manifold Sets:

- Up to 800 PSIG High side
- Up to 250 PSIG Low Side
- 550 PSIG Low Side Retard

### Manifold Hoses:

- Service Pressure Rating of 800 PSIG

### Recovery Cylinders:

- 400 PSIG Pressure Rating
- Dept. of Transportation 4BA400 or BW400

### CAUTION

*R-410A systems operate at higher pressures than R-22 systems. Do not use R-22 service equipment or components on R-410A equipment.*

## SPECIFICATION OF R-410A:

**Application:** R-410A is not a drop-in replacement for R-22; equipment designs must accommodate its higher pressures. It cannot be retrofitted into R-22 condensing units.



**Physical Properties:** R-410A has an atmospheric boiling point of -62.9°F and its saturation pressure at 77°F is 224.5 psig.

**Composition:** R-410A is an azeotropic mixture of 50% by weight difluoromethane (HFC-32) and 50% by weight pentafluoroethane (HFC-125).

**Pressure:** The pressure of R-410A is approximately 60% (1.6 times) greater than R-22. Recovery and recycle equipment, pumps, hoses and the like need to have design pressure ratings appropriate for R-410A. *Manifold sets need to range up to 800 psig high-side and 250 psig low-side with a 550 psig low-side retard. Hoses need to have a service pressure rating of 800 psig. Recovery cylinders need to have a 400 psig service pressure rating.* DOT 4BA400 or DOT BW400.

**Combustibility:** At pressures above 1 atmosphere, mixture of R-410A and air can become combustible. R-410A and air should never be mixed in tanks or supply lines, or be allowed to accumulate in storage tanks. Leak checking should never be done with a mixture of R-410A and air. Leak checking can be performed safely with nitrogen or a mixture of R-410A and nitrogen.

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## QUICK REFERENCE GUIDE FOR R-410A

- R-410A refrigerant operates at approximately 60% higher pressure (1.6 times) than R-22. Ensure that servicing equipment is designed to operate with R-410A.
- R-410A refrigerant cylinders are pink in color.
- R-410A, as with other HFC's is only compatible with POE oils.
- Vacuum pumps will not remove moisture from oil.
- R-410A systems are to be charged with liquid refrigerants. Prior to March 1999, R-410A refrigerant cylinders had a dip tube. These cylinders should be kept upright for equipment charging. Post March 1999 cylinders do not have a dip tube and should be inverted to ensure liquid charging of the equipment.
- Do not install a suction line filter drier in the liquid line.
- A liquid line filter drier is standard on every unit. Only manufacturer approved liquid line filter driers can be used. These are Sporlan (CW083S) and Alco (80K083S) driers. These filter driers are rated for minimum working pressure of 600 psig.
- Desiccant (drying agent) must be compatible for POE oils and R-410A.

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## REPLACEMENT UNITS

To prevent failure of a new condensing unit, the existing evaporator tubing system must be correctly sized and cleaned or replaced. Care must be exercised that the expansion device is not plugged. For new and replacement units, a liquid line filter drier should be installed and refrigerant tubing should be properly sized. Test the oil for acid. If positive, a suction line filter drier is mandatory.

**IMPORTANT:** WHEN REPLACING AN R-22 UNIT WITH AN R-410A UNIT, EITHER REPLACE THE LINE SET OR ENSURE THAT THE EXISTING LINE SET IS THOROUGHLY CLEANED OF ANY OLD OIL OR DEBRIS.

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## EVAPORATOR COIL

**REFER TO EVAPORATOR COIL MANUFACTURER'S INSTALLATION INSTRUCTIONS.**

**IMPORTANT:** The manufacturer is not responsible for the performance and operation of a mismatched system, or for a match listed with another manufacturer's coil.

### CAUTION

Only use evaporators approved for use on R-410A systems. Use of existing R-22 evaporators can introduce mineral oil to the R-410A refrigerant forming two different liquids and decreasing oil return to the compressor. This can result in compressor failure.

**NOTE: All (-)ANL & (-)APM units must be installed with a TEV Evaporator.**

The thermostatic expansion valve is specifically designed to operate with R-410A. **DO NOT use an R-22 TEV or evaporator. The existing evaporator must be replaced with the factory specified TEV evaporator specifically designed for R-410A.**

## LOCATION

Do not install the indoor evaporator coil in the return duct system of a gas or oil furnace. Provide a service inlet to the coil for inspection and cleaning. Keep the coil pitched toward the drain connection.

### CAUTION

When coil is installed over a finished ceiling and/or living area, it is recommended that a secondary sheet metal condensate pan be constructed and installed under entire unit. Failure to do so can result in property damage.

## INTERCONNECTING TUBING

### VAPOR AND LIQUID LINES

Keep all lines sealed until connection is made.

Make connections at the indoor coil first.

Refer to Line Size Information in Tables 3 and 4 for correct size and multipliers to be used to determine capacity for various vapor line diameters and lengths of run. The losses due to the lines being exposed to outdoor conditions are not included.

The factory refrigeration charge in the outdoor unit is sufficient for 15 feet of interconnecting lines. The factory refrigeration charge in the outdoor unit is sufficient for the unit and 15 feet of standard size interconnecting liquid and vapor lines. For different lengths, adjust the charge as indicated below.

1/4" ± 0.3 oz. per foot

5/16" ± 0.4 oz. per foot

3/8" ± 0.6 oz. per foot

1/2" ± 1.2 oz. per foot

### MAXIMUM LENGTH OF LINES

The maximum length of interconnecting line is 150 feet. Always use the shortest length possible with a minimum number of bends. Additional compressor oil is not required for any length up to 150 feet.

**NOTE:** Excessively long refrigerant lines cause loss of equipment capacity.

### OUTDOOR UNIT INSTALLED ABOVE INDOOR COIL

Keep the vertical separation between coils to a minimum. However, the vertical distance can be as great as 120 feet with the condensing unit ABOVE the indoor coil. Use the following guidelines when installing the unit:

1. DO NOT exceed 120 feet maximum vertical separation.
2. DO NOT change the flow check piston sizes if the vertical separation does not exceed the values in Table 4.
3. Flow Check Piston Coil:
  - a. The vertical separation can be greater than the value in Table 6, but no more than 120 feet.
  - b. If the separation height exceeds the Table value, reduce the indoor coil flow check piston by two sizes plus one size for additional 10 feet beyond the Table value.
4. Expansion Valve Coil:
  - a. The vertical separation can be greater than the Table value, but no more than 120 feet.
  - b. No changes are required for expansion valve coils.

5. Capillary Tube Coil:  
DO NOT exceed the Table values for vertical separation for capillary tube coils.
6. Always use the smallest liquid line size permitted to minimize the system charge.
7. Table 5 may be used for sizing horizontal runs.

## **OUTDOOR UNIT BELOW INDOOR COIL**

Keep the vertical separation to a minimum. Use the following guidelines when installing the unit:

1. DO NOT exceed the vertical separations as indicated on Table 4.
2. Always use the smallest liquid line size permitted to minimize system charge.
3. No changes are required for either flow check piston coils or expansions coils.
4. Table 4 may be used for sizing horizontal runs.

## **TUBING INSTALLATION**

Observe the following when installing correctly sized type “L” refrigerant tubing between the condensing unit and evaporator coil:

- If a portion of the liquid line passes through a hot area where liquid refrigerant can be heated to form vapor, insulating the liquid line is required.
- Use clean, dehydrated, sealed refrigeration grade tubing.
- Always keep tubing sealed until tubing is in place and connections are to be made.
- Blow out the liquid and vapor lines with dry nitrogen before connecting to the outdoor unit and indoor coil. Any debris in the line set will end up plugging the expansion device.
- As an added precaution, a high quality filter drier is standard on R-410A units.
- Do not allow the vapor line and liquid line to be in contact with each other. This causes an undesirable heat transfer resulting in capacity loss and increased power consumption. The vapor line must be insulated.
- If tubing has been cut, make sure ends are deburred while holding in a position to prevent chips from falling into tubing. Burrs such as those caused by tubing cutters can affect performance dramatically, particularly on small liquid line sizes.
- For best operation, keep tubing run as short as possible with a minimum number of elbows or bends.
- Locations where the tubing will be exposed to mechanical damage should be avoided. If it is necessary to use such locations, the copper tubing should be housed to prevent damage.
- If tubing is to be run underground, it must be run in a sealed watertight chase.
- Use care in routing tubing and do not kink or twist. Use a good tubing bender on the vapor line to prevent kinking.
- Route the tubing using temporary hangers, then straighten the tubing and install permanent hangers. Line must be adequately supported.
- The vapor line must be insulated to prevent dripping (sweating) and prevent performance losses. Armaflex and Rubatex are satisfactory insulations for this purpose. Use 1/2” minimum insulation thickness, additional insulation may be required for long runs.
- Check Table 3 for the correct vapor line size. Check Table 4 for the correct liquid line size.

**TABLE 3**  
**SUCTION LINE LENGTH/SIZE VS. CAPACITY MULTIPLIER (R410A)**

| UNIT SIZE                    |          | 1½ Ton         | 2 Ton          | 2½ Ton         | 3 Ton          | 3½ Ton           | 4 Ton            | 5 Ton            |
|------------------------------|----------|----------------|----------------|----------------|----------------|------------------|------------------|------------------|
| Suction Line Connection Size |          | 3/4" [19] I.D. | 3/4" [19] I.D. | 3/4" [19] I.D. | 3/4" [19] I.D. | 7/8" [22.2] I.D. | 7/8" [22.2] I.D. | 7/8" [22.2] I.D. |
| Suction Line Run - Feet [m]  |          | 5/8" [15.9]    | 5/8" [15.9]    | 5/8" [15.9]    | —              | 3/4" [19]        | —                | —                |
|                              |          | 3/4" [19]*     | 3/4" [19]*     | 3/4" [19]*     | 3/4" [19]*     | 7/8" [22.2]*     | 7/8" [22.2]*     | 7/8" [22.2]*     |
|                              |          | —              | —              | 7/8            | —              | —                | —                | —                |
| 25'<br>[7.6]                 | Optional | 1.00           | 1.00           | 1.00           | —              | 1.00             | —                | —                |
|                              | Standard | 1.00           | 1.00           | 1.00           | 1.00           | 1.00             | 1.00             | 1.00             |
|                              | Optional | —              | —              | 1.00           | 1.00           | —                | 1.00             | 1.00             |
| 50'<br>[15.24]               | Optional | 0.98           | 0.98           | 0.96           | —              | 0.99             | —                | —                |
|                              | Standard | 0.99           | 0.99           | 0.98           | 0.98           | 0.99             | 0.99             | 0.99             |
|                              | Optional | —              | —              | 0.99           | 0.99           | —                | 0.99             | 0.99             |
| 100'<br>[30.5]               | Optional | 0.95           | 0.95           | 0.94           | —              | 0.96             | —                | —                |
|                              | Standard | 0.96           | 0.96           | 0.96           | 0.96           | 0.98             | 0.96             | 0.97             |
|                              | Optional | —              | —              | 0.97           | 0.97           | —                | 0.98             | 0.98             |
| 150'<br>[45.7]               | Optional | 0.92           | 0.92           | 0.91           | —              | 0.94             | —                | —                |
|                              | Standard | 0.93           | 0.94           | 0.93           | 0.94           | 0.96             | 0.95             | 0.94             |
|                              | Optional | —              | —              | 0.95           | 0.95           | —                | 0.96             | 0.97             |

**NOTES:**

\*Standard

Note: Using suction line larger than shown in chart will result in poor oil return and is not recommended.

[ ] Designates Metric Conversions

## TUBING CONNECTIONS

Indoor evaporator coils have only a holding charge of dry nitrogen. Keep all tube ends sealed until connections are to be made.

- Use type "L" copper refrigeration tubing. Braze the connections with the following alloys:
  - copper to copper - 5%
  - Silver alloy (no flux)
  - copper to steel or brass - 35%
  - silver alloy (with flux)
- Be certain both refrigerant shutoff valves at the outdoor unit are closed.
- Clean the inside of the fittings and outside of the tubing with steel wool or sand cloth before soldering. Always keep chips, steel wool, dirt, etc., out of the inside when cleaning.
- Assemble tubing part way into fitting. Apply flux all around the outside of the tubing and push tubing into stop. This procedure will keep the flux from getting inside the system.
- Remove the cap and schrader core from service port to protect seals from heat damage.
- Use an appropriate heatsink material around the copper stub and the service valves before applying heat.
- **IMPORTANT:** Do not braze any fitting with the TEV sensing bulb attached.
- Braze the tubing between the outdoor unit and indoor coil. Flow dry nitrogen into a service port and through the tubing while brazing.
- After brazing – use an appropriate heatsink material to cool the joint and remove any flux residue.
- The service valves are not backseating valves. To open the valves, remove the valve cap with an adjustable wrench. Insert a 3/16" or 5/16" hex wrench into the stem. Back out counterclockwise.
- Replace the valve cap finger tight then tighten an additional 1/2 hex flat for a metal-to-metal seal.

**TABLE 4**  
LIQUID LINE SIZE — OUTDOOR UNIT ABOVE INDOOR COIL

| System Capacity | Line Size Connection Size (Inch I.D.) [mm] | Line Size (Inch I.D.) [mm] | Liquid Line Size<br>Outdoor Unit Above Indoor Coil<br>(Cooling Only - Does not apply to Heat Pumps) |           |           |            |            |            |
|-----------------|--------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------|-----------|-----------|------------|------------|------------|
|                 |                                            |                            | Total                                                                                               |           |           |            |            |            |
|                 |                                            |                            | 25 [7.6]                                                                                            | 50 [15.2] | 75 [22.9] | 100 [30.5] | 125 [38.1] | 150 [45.7] |
|                 |                                            |                            | Minimum Vertical Separation - Feet [m] ("0" means no restriction)                                   |           |           |            |            |            |
| 1½ Ton          | 3/8" [9.5]                                 | 1/4 [6.4]                  | 0                                                                                                   | 0         | 0         | 0          | 8 [2.4]    | 24 [7.3]   |
|                 |                                            | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 2 Ton           | 3/8" [9.5]                                 | 1/4 [6.4]                  | 0                                                                                                   | 3 [0.9]   | 29 [8.8]  | 55 [16.8]  | 81 [24.7]  | 108 [32.9] |
|                 |                                            | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 2½ Ton          | 3/8" [9.5]                                 | 1/4 [6.4]                  | 0                                                                                                   | 14 [4.3]  | 56 [17.1] | 98 [29.9]  | N/A        | N/A        |
|                 |                                            | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 3 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 9          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 3½ Ton          | 3/8" [9.5]                                 | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 16 [4.9]   | 35 [10.7]  | 54 [16.5]  |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 4 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| 5 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
|                 |                                            | 3/8* [9.5]                 | 0                                                                                                   | 0         | 0         | 0          | 0          | 0          |
| System Capacity | Line Size Connection Size (Inch I.D.) [mm] | Line Size (Inch I.D.) [mm] | Liquid Line Size<br>Outdoor Unit Below Indoor Coil                                                  |           |           |            |            |            |
|                 |                                            |                            | Total                                                                                               |           |           |            |            |            |
|                 |                                            |                            | 25 [7.6]                                                                                            | 50 [15.2] | 75 [22.9] | 100 [30.5] | 125 [38.1] | 150 [45.7] |
|                 |                                            |                            | Maximum Vertical Separation - Feet [m]                                                              |           |           |            |            |            |
| 1½ Ton          | 3/8" [9.5]                                 | 1/4 [6.4]                  | 25 [7.6]                                                                                            | 40 [12.2] | 25 [7.6]  | 9 [2.7]    | N/A        | N/A        |
|                 |                                            | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 50 [15.2] | 62 [18.9] | 58 [17.7]  | 53 [16.2]  | 49 [14.9]  |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 75 [22.9] | 72 [21.9]  | 70 [21.3]  | 68 [20.7]  |
| 2 Ton           | 3/8" [9.5]                                 | 1/4 [6.4]                  | 25 [7.6]                                                                                            | N/A       | N/A       | N/A        | N/A        | N/A        |
|                 |                                            | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 36 [10.9] | 29 [8.8]  | 23 [7.0]   | 16 [4.9]   | 9 [2.7]    |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 72 [21.9] | 70 [21.3]  | 68 [20.7]  | 65 [19.8]  |
| 2½ Ton          | 3/8" [9.5]                                 | 1/4 [6.4]                  | 25 [7.6]                                                                                            | N/A       | N/A       | N/A        | N/A        | N/A        |
|                 |                                            | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 49 [14.9] | 38 [11.6] | 27 [8.3]   | 17 [5.2]   | 6 [1.8]    |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 68 [20.7] | 85 [19.8]  | 62 [18.9]  | 58 [17.7]  |
| 3 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 50 [15.2] | 37 [11.3] | 22 [6.7]   | 7 [2.1]    | N/A        |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 68 [20.7] | 63 [19.2]  | 58 [17.7]  | 53 [16.2]  |
| 3½ Ton          | 3/8" [9.5]                                 | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 27 [7.0]  | 4 [1.2]   | N/A        | NA         | N/A        |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 43 [13.1] | 36 [10.9]  | 30 [9.1]   | 24 [7.3]   |
| 4 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 46 [14.0] | 38 [11.6] | 30 [9.1]   | 22 [6.7]   | 15 [4.5]   |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 56 [17.1] | 55 [16.8]  | 53 [16.2]  | 52 [15.8]  |
| 5 Ton           | 3/8" [9.5]                                 | 5/16 [7.9]                 | 25 [7.6]                                                                                            | 50 [15.2] | 56 [17.1] | 44 [13.4]  | 32 [9.8]   | 20 [6.1]   |
|                 |                                            | 3/8* [9.5]                 | 25 [7.6]                                                                                            | 50 [15.2] | 75 [22.9] | 81 [24.7]  | 79 [24.1]  | 76 [23.2]  |

**NOTES:**

\*Standard

N/A = Application not recommended.

[ ] Designates Metric Conversions

## LEAK TESTING

- Pressurize line set and coil through service fittings with dry nitrogen to 150 psig maximum. Leak test all joints using liquid detergent. If a leak is found, repair and repeat leak test procedures.

### WARNING

**DO NOT USE OXYGEN TO PURGE LINES OR PRESSURIZE SYSTEM FOR LEAK TEST. OXYGEN REACTS VIOLENTLY WITH OIL, WHICH CAN CAUSE AN EXPLOSION RESULTING IN SEVERE PERSONAL INJURY OR DEATH.**

## START-UP – CHECKING AIRFLOW

The air distribution system has the greatest effect on airflow. The duct system is totally controlled by the contractor. For this reason, the contractor should use only industry-recognized procedures. The correct air quantity is critical to air conditioning systems. Proper operation, efficiency, compressor life, and humidity control depend on the correct balance between indoor load and outdoor unit capacity. Excessive indoor airflow increases the possibility of high humidity problems. Low indoor airflow reduces total capacity and causes coil icing. Serious harm can be done to the compressor by low airflow, such as that caused by refrigerant flooding. Each ton of cooling requires between 375 and 450 cubic feet of air per minute (CFM). See the manufacturer's spec sheet for rated airflow for the system being installed. Duct design and construction should be carefully done. System performance can be lowered dramatically through bad planning or workmanship. Air supply diffusers must be selected and located carefully. They must be sized and positioned to deliver treated air along the perimeter of the space. If they are too small for their intended airflow, they become noisy. If they are not located properly, they cause drafts. Return air grilles must be properly sized to carry air back to the blower. If they are too small, they also cause noise. The installers should balance the air distribution system to ensure proper quiet airflow to all rooms in the home. This ensures a comfortable living space.

These simple mathematical formulas can be used to determine the CFM in a residential or light commercial system. Electric resistance heaters can use:

$$CFM = \frac{\text{volts} \times \text{amps} \times 3.413}{SHC \times \text{temp rise}}$$

Gas furnaces can use:

$$CFM = \frac{\text{Output Capacity in BTUH}^*}{SHC \times \text{temp rise}}$$

\*Refer to furnace data plate for furnace output capacity. SHC = Sensible Heat Constant (see table below), an air velocity meter or airflow hood can give a more accurate reading of the system CFM. The measurement for temperature rise should be performed at the indoor coil inlet and near the outlet, but out of direct line of sight of the heater element or heat exchanger. For best results, measure air temperature at multiple points and average the measurements to obtain coil inlet and outlet temperatures.

| Altitude<br>(feet) | SENSIBLE HEAT<br>CONSTANT<br>(SHC) | ALTITUDE<br>(FEET) | SENSIBLE HEAT<br>CONSTANT<br>(SHC) |
|--------------------|------------------------------------|--------------------|------------------------------------|
| Sea Level          | 1.08                               | 6000               | 0.87                               |
| 500                | 1.07                               | 7000               | 0.84                               |
| 1000               | 1.05                               | 8000               | 0.81                               |
| 2000               | 1.01                               | 9000               | 0.78                               |
| 3000               | 0.97                               | 10000              | 0.75                               |
| 4000               | 0.94                               | 15000              | 0.61                               |
| 5000               | 0.90                               | 20000              | 0.50                               |

## EVACUATION AND LEAK TESTING

### EVACUATION PROCEDURE

Evacuation is the most important part of the entire service procedure. The life and efficiency of the equipment is dependent upon the thoroughness exercised by the serviceman when evacuating air and moisture from the system.

Air or nitrogen in the system causes high condensing temperatures and pressure, resulting in increased power input and non-verifiable performance.

Moisture chemically reacts with the refrigerant and oil to form corrosive hydrofluoric acid. This attacks motor windings and parts, causing breakdown.

- After the system has been leak-checked and proven sealed, connect the vacuum pump and evacuate system to 500 microns and hold 500 microns or less for at least 15 minutes. The vacuum pump must be connected to both the high and low sides of the system by connecting to the two pressure ports. Use the largest size connections available since restrictive service connections may lead to false readings because of pressure drop through the fittings.
- After adequate evacuation, open both service valves by removing both brass service valve caps with an adjustable wrench. Insert a 3/16" [5 mm] or 5/16" [8 mm] hex wrench into the stem and turn counterclockwise until the wrench stops.
- At this time gauges must be connected to the access fitting on the liquid line (small) service valve and the common suction port connected to the common suction line between the reversing valve and compressor to check and adjust charge.

**IMPORTANT:** Compressors (especially scroll type) should never be used to evacuate the air conditioning system because internal electrical arcing may result in a damaged or failed compressor. Never run a scroll compressor while the system is in a vacuum or compressor failure will occur.

### FINAL LEAK TESTING

After the unit has been properly evacuated and service valves opened, a halogen leak detector should be used to detect leaks in the system. If a leak is detected, the refrigerant should be recovered before repairing the leak. The Clean Air Act prohibits releasing refrigerant into the atmosphere.

## CHECKING REFRIGERANT CHARGE

### WARNING

**The top of the scroll compressor shell is hot. Touching the compressor top may result in serious personal injury.**

Charge for all systems should be checked against the Charging Chart inside the access panel cover.

**IMPORTANT:** Use factory-approved charging method as outlined on the next 4 pages to ensure proper system charge.



## NOTICE

The optimum refrigerant charge for any outdoor unit matched with a CFL/CFM/H\*L indoor coil/air handler is affected by the application. Therefore, charging data has been developed to assist the field technician in optimizing the charge for all mounting configurations. Refer to the charging chart inside the access panel cover on the unit and choose the appropriate column for the specific application being installed or serviced.

## CHARGING UNITS WITH R-410A REFRIGERANT

### CAUTION

R-410A pressures are approximately 60% higher (1.6 times) than R-22 pressures. Use appropriate care when using this refrigerant. Failure to exercise care may result in equipment damage or personal injury.

Charge for all systems should be checked against the Charging Chart inside the access panel cover.

**IMPORTANT:** Do not operate the compressor without charge in the system.

Addition of R-410A will raise high-side pressures (liquid, and discharge).

### NOTICE

System maintenance is to be performed by a qualified and certified technician.

The following method is used for charging systems in the cooling and heating mode. All steps listed should be performed to insure proper charge has been set. For measuring pressures, the service valve port on the liquid valve (small valve) and the service port on the vapor valve (large valve) are to be used.

#### CONFIRM ID AIR FLOW & COILS ARE CLEAN

Confirm adequate Indoor supply air flow prior to starting the system. See the Technical Specification sheet for rated air flow for each ID/OD unit match. Air filter(s) and coils (indoor & outdoor) are to be clean and free of frost prior to starting the system. Supply Air flow must be between 375 and 450 cfm per rated cooling ton prior to adjusting system charge. If a humidification system is installed disengage it from operating prior to charge adjustment. Refer to the "Checking Airflow" section of this manual for further instruction.

### NOTICE

Verify system components are matched according to the outdoor unit Specification Sheet.

## MEASUREMENT DEVICE SETUP

- Step 1. With an R410A gauge set, attach the high pressure hose to the access fitting on the liquid line (small) service valve at the OD unit.
- Step 2. Attach the low pressure hose to the access fitting on the vapor line (large) service valve at the OD unit.
- Step 3. Attach a temperature probe within 6" outside of the unit on the copper liquid line (small line). For more accurate measurements clean the copper line prior to measurement and use a calibrated clamp on temperature probe or an insulated surface thermocouple.

## CHARGING BY WEIGHT

### NOTICE

**ADJUST THE SYSTEM CHARGE BY WEIGHT FOR THE STRAIGHT LENGTH OF THE REFRIGERANT LINE SET.**

For a new installation, evacuation of interconnecting tubing and indoor coil is adequate; otherwise, evacuate the entire system. Use the factory charge shown in “Electrical and Physical Data” on page 4 of these instructions or on the unit data plate. Note that the charge value includes charge required for 15 ft. [4.6 m] of standard-size inter-connecting liquid line without a filter drier. Calculate actual charge required with installed liquid line size and length using:

1/4" [6.4 mm] O.D. = .3 oz./ft. [8.5 g/.30 m]

5/16" [7.9 mm] O.D. = .4 oz./ft. [11.3 g/.30 m]

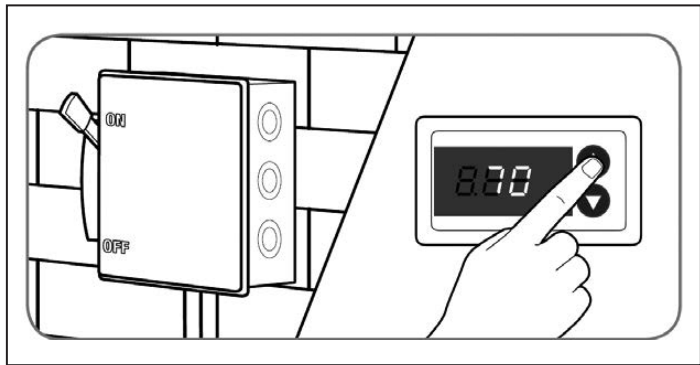
3/8" [9.5 mm] O.D. = .6 oz./ft. [17.0 g/.30 m]

1/2" [12.7 mm] O.D. = 1.2 oz./ft. [34.0 g/.30 m]

Add 6 oz. for field-installed filter drier.

With an accurate scale (+/- 1 oz. [28.3 g]) or volumetric charging device, adjust charge difference between that shown on the unit data plate and that calculated for the new system installation. If the entire system has been evacuated, add the total calculated charge.

**IMPORTANT:** Charging by weight is not always accurate since the application can affect the optimum refrigerant charge. Charging by weight is considered a starting point ONLY. Always check the charge by using the charging chart and adjust as necessary. CHARGING BY LIQUID SUB-COOLING MUST BE USED FOR FINAL CHARGE ADJUSTMENT.



With thermostat in the “Off” position, turn the power on to the furnace or air handler. Start the furnace or air handler with the thermostat.

## GROSS CHARGING BY PRESSURES

Step 1. Following air flow verification and charge weigh in, run the unit for a minimum of 15 minutes prior to noting pressures and temperature.

**IMPORTANT:** Indoor conditions as measured at the indoor coil must be within 2°F of the following during gross charge (pressure) evaluation:

Cooling Mode: 80°F Dry Bulb

### NOTICE

**If the Indoor temperature is above or below this range, run the system to bring the temperature down or run the electric heat/furnace to bring the temperature within this range. System pressure values provided in the Charge Chart for outdoor dry bulbs corresponding to conditions outside of ranges listed below, are provided as reference ONLY.**

Step 2. Note the Outdoor Dry Bulb Temperature, ODDB°F = \_\_\_\_\_°F. Unit charging is recommended under the following outdoor conditions ONLY:

Cooling Mode ONLY: 55°F outdoor dry bulb and above

- Step 3. Locate and note the design pressures. The correct liquid and vapor pressures are found at the intersection of the Installed system and the outdoor ambient temperature on the Charging Chart located on the inside of the control box cover of the outdoor unit.

Liquid Pressure: = \_\_\_\_\_psig; Vapor Pressure = \_\_\_\_\_psig

### NOTICE

**The refrigerant pressures provided are for gross charge check ONLY. These pressure values are typical, but may vary due to application. Evaporator load will cause pressures to deviate. Notice that all systems have unique pressure curves. The variation in the slope and value is determined by the component selection for that indoor/outdoor matched system. The variation from system to system seen in the table is normal.**

- Step 4. If the measured liquid pressure is below the listed requirement for the given outdoor and indoor conditions, add charge. If the measured liquid pressure is above the listed requirement for the given Outdoor and Indoor conditions remove charge.

### FINAL CHARGE BY SUB-COOLING

- Step 1. After gross charging note the designed Sub-Cool value. The correct sub-cooling value is found at the intersection of the Installed system and the outdoor ambient temperature on the Charging Chart located on the inside of the control box cover of the outdoor unit.

SC° from Charging Chart = \_\_\_\_\_°F.

**IMPORTANT:** Indoor conditions as measured at the indoor coil are required to be between 70°F and 80°F dry bulb for fine tune unit charge adjustment. Unit charging is recommended under the following outdoor conditions ONLY:

Cooling Mode ONLY: 55°F outdoor dry bulb and above

### NOTICE

**If the Indoor temperature is above or below the recommended range, run the system to bring the temperature down or run the electric heat/furnace to bring the temperature up. System sub-cooling values provided in the Charge Chart for outdoor dry bulbs corresponding to conditions outside of the above range, are provided as reference ONLY.**

- Step 2. Note the measured Liquid Pressure, Pliq = \_\_\_\_\_psig, as measured from the liquid (small) service valve. Use the pressure temperature chart on the next page to note the corresponding saturation temperature for R410A at the measured liquid pressure.

Liquid Saturation Temperature, SAT°F= \_\_\_\_\_°F.

- Step 3. Note the liquid line temperature, Liq° = \_\_\_\_\_°F, as measured from a temperature probe located within 6" outside of the unit on the copper liquid line (small line). It is recommended to use a calibrated clamp on temperature probe or an insulated surface thermocouple.

- Step 4. Subtract the liquid line temperature (Step 3) from the saturation temperature (Step 2) to calculate Sub-Cooling. SAT°F \_\_\_\_\_ - Liq°F \_\_\_\_\_ = SC°F \_\_\_\_\_

- Step 5. Adjust Charge to obtain the specified sub-cooling value. If the measured sub-cool is below the listed requirement for the given outdoor and indoor conditions, add charge. If the measured sub-cool is above the listed requirement for the given outdoor and indoor conditions remove charge.

**IMPORTANT:** Excessive use of elbows in the refrigerant line set can produce excessive pressure drop. Follow industry best practices for installation. Installation and commissioning of this equipment is to be preformed by trained and qualified HVAC professionals. For technical assistance contact your Distributor Service Coordinator.

TABLE 5

| SATURATION<br>TEMP<br>(Deg. F) | R-410A<br>PSIG | SATURATION<br>TEMP<br>(Deg. F) | R-410A<br>PSIG | SATURATION<br>TEMP<br>(Deg. F) | R-410A<br>PSIG | SATURATION<br>TEMP<br>(Deg. F) | R-410A<br>PSIG |
|--------------------------------|----------------|--------------------------------|----------------|--------------------------------|----------------|--------------------------------|----------------|
| -150                           | -              | -30                            | 17.9           | 35                             | 107.5          | 100                            | 317.4          |
| -140                           | -              | -25                            | 22             | 40                             | 118.5          | 105                            | 340.6          |
| -130                           | -              | -20                            | 26.4           | 45                             | 130.2          | 110                            | 365.1          |
| -120                           | -              | -15                            | 31.3           | 50                             | 142.7          | 115                            | 390.9          |
| -110                           | -              | -10                            | 36.5           | 55                             | 156.0          | 120                            | 418.0          |
| -100                           | -              | -5                             | 42.2           | 60                             | 170.1          | 125                            | 446.5          |
| -90                            | -              | 0                              | 48.4           | 65                             | 185.1          | 130                            | 476.5          |
| -80                            | -              | 5                              | 55.1           | 70                             | 201.0          | 135                            | 508.0          |
| -70                            | -              | 10                             | 62.4           | 75                             | 217.8          | 140                            | 541.2          |
| -60                            | 0.4            | 15                             | 70.2           | 80                             | 235.6          | 145                            | 576.0          |
| -50                            | 5.1            | 20                             | 78.5           | 85                             | 254.5          | 150                            | 612.8          |
| -40                            | 10.9           | 25                             | 87.5           | 90                             | 274.3          |                                |                |
| -35                            | 14.2           | 30                             | 97.2           | 95                             | 295.3          |                                |                |

## FINISHING UP INSTALLATION

- Disconnect pressure gauges from pressure ports; then replace the pressure port caps and tighten adequately to seal caps. **Do not over tighten.**
- Replace the service valve caps finger-tight and then tighten with an open-end wrench adequately to seal caps. **Do not over tighten.**
- Replace control box cover and service panel and install screws to secure service panel.
- Restore power to unit at disconnect if required.
- Configure indoor thermostat per the thermostat installation instructions and set thermostat to desired mode and temperature.

## WARNING

TURN OFF ELECTRIC POWER AT THE FUSE BOX OR SERVICE PANEL BEFORE MAKING ANY ELECTRICAL CONNECTIONS.

ALSO, THE GROUND CONNECTION MUST BE COMPLETED BEFORE MAKING LINE VOLTAGE CONNECTIONS. FAILURE TO DO SO CAN RESULT IN ELECTRICAL SHOCK, SEVERE PERSONAL INJURY OR DEATH.

## ELECTRICAL WIRING

Field wiring must comply with the National Electric Code (C.E.C. in Canada) and any applicable local code.

### POWER WIRING

It is important that proper electrical power from a commercial utility is available at the condensing unit contactor. Voltage ranges for operation are shown in Table 6.

Install a branch circuit disconnect within sight of the unit and of adequate size to handle the starting current (see Tables 1 and 2).

Power wiring must be run in a rain-tight conduit. Conduit must be run through the connector panel below the access cover (see Figure 1) and attached to the bottom of the control box.

Connect power wiring to contactor located in outdoor condensing unit electrical box. (See wiring diagram attached to unit access panel.)

Check all electrical connections, including factory wiring within the unit and make sure all connections are tight.

DO NOT connect aluminum field wire to the contactor terminals.

**TABLE 6**  
VOLTAGE RANGES (60 HZ)

| Nameplate Voltage | Operating Voltage Range at Copeland<br>Maximum Load Design Conditions for<br>Compressors |
|-------------------|------------------------------------------------------------------------------------------|
| 208/230 (1 Phase) | 187 - 253                                                                                |
| 208/230 (3 Phase) | 187 - 253                                                                                |
| 460               | 414 - 506                                                                                |
| 575               | 517 - 633                                                                                |

## GROUNDING

A grounding lug is provided near the contactor for a ground wire.

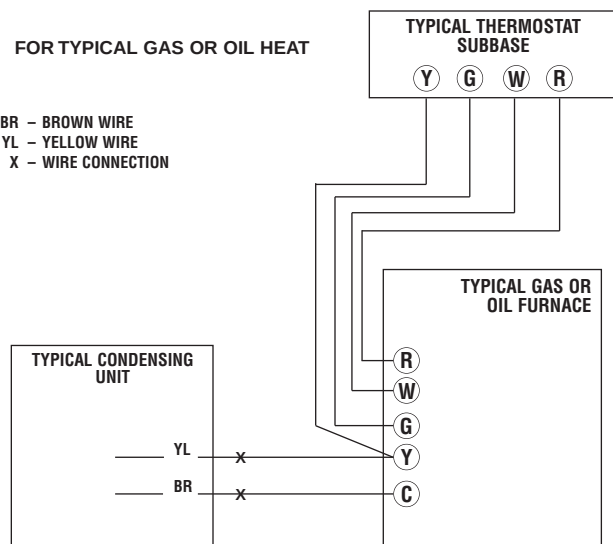
### **⚠ WARNING**

**THE UNIT MUST BE PERMANENTLY GROUNDED. FAILURE TO DO SO CAN CAUSE ELECTRICAL SHOCK RESULTING IN SEVERE PERSONAL INJURY OR DEATH.**

**FIGURE 2**  
CONTROL WIRING FOR GAS OR OIL FURNACE

FOR TYPICAL GAS OR OIL HEAT

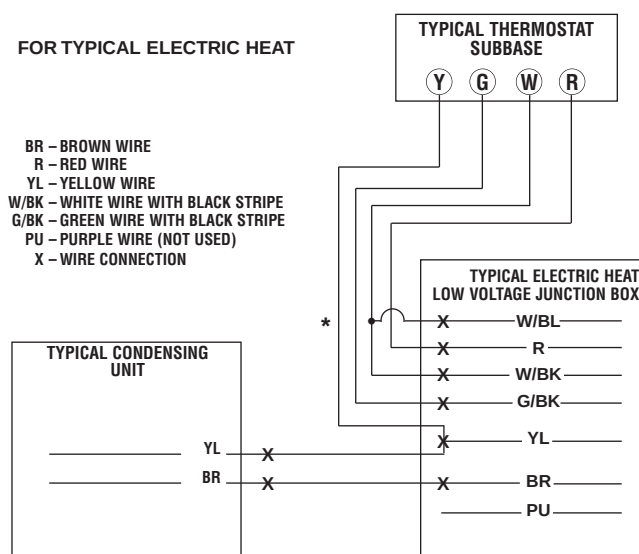
BR - BROWN WIRE  
YL - YELLOW WIRE  
X - WIRE CONNECTION



\*IF MAXIMUM OUTLET TEMPERATURE RISE IS DESIRED, IT IS RECOMMENDED THAT W1 (W/BK) AND W2 (W/BL) BE JUMPED TOGETHER.

FOR TYPICAL ELECTRIC HEAT

BR - BROWN WIRE  
R - RED WIRE  
YL - YELLOW WIRE  
W/BK - WHITE WIRE WITH BLACK STRIPE  
G/BK - GREEN WIRE WITH BLACK STRIPE  
PU - PURPLE WIRE (NOT USED)  
X - WIRE CONNECTION



**TABLE 7**  
FIELD WIRE SIZE FOR 24 VOLT THERMOSTAT CIRCUITS

| Thermostat Load - Amps   | SOLID COPPER WIRE - AWG. |    |     |     |     |     |
|--------------------------|--------------------------|----|-----|-----|-----|-----|
|                          | 3.0                      | 16 | 14  | 12  | 10  | 10  |
|                          | 2.5                      | 16 | 14  | 12  | 12  | 10  |
| Thermostat Load - Amps   | 2.0                      | 18 | 16  | 14  | 12  | 10  |
|                          |                          | 50 | 100 | 150 | 200 | 250 |
| Length of Run - Feet (1) |                          |    |     |     |     |     |
|                          |                          |    |     | 300 |     |     |

(1) Wire length equals twice the run distance.

NOTE: Do not use control wiring smaller than No. 18 AWG between thermostat and outdoor unit.

## CONTROL WIRING

(See Figure 2)

If the low voltage control wiring is run in conduit with the power supply, Class I insulation is required. Class II insulation is required if run separate. Low voltage wiring may be run through the insulated bushing provided in the 7/8 hole in the base panel, up to and attached to the pigtails from the bottom of the control box. Conduit can be run to the base panel if desired by removing the insulated bushing.

A thermostat and a 24 volt, 40 VA minimum transformer are required for the control circuit of the condensing unit. The furnace or the air handler transformer may be used if sufficient. See the wiring diagram for reference. Use Table 8 to size the 24 volt control wiring.

## HIGH AND LOW PRESSURE CONTROLS (HPC OR LPC)

These controls keep the compressor from operating in pressure ranges which can cause damage to the compressor. Both controls are in the low voltage control circuit.

High pressure control (HPC) is a manual reset which opens near 610 PSIG. Do not reset arbitrarily without first determining what caused it to trip.

The low pressure control (LPC) is an automatic reset which opens near 50 PSIG and closes near 95 PSIG.

**NOTE:** High and low pressure controls are standard on all (-)ANL, (-)APM models.

## FIELD INSTALLED ACCESSORIES

### COMPRESSOR CRANKCASE HEAT (CCH)

While scroll compressors usually do not require crankcase heaters, there are instances when a heater should be added. Refrigerant migration during the off cycle can result in a noisy start up. Add a crankcase heater to minimize refrigeration migration, and to help eliminate any start up noise or bearing "wash out."

**NOTE:** A crankcase heater should be installed if: the charge of the system exceeds the values in Tables 4 and 5, if the system is subject to voltage variations or when a low ambient control is used for system operation below 55°F.

All heaters are located on the lower half of the compressor shell. Its purpose is to drive refrigerant from the compressor shell during long off cycles, thus preventing damage to the compressor during start-up.

At initial start-up or after extended shutdown periods, make sure the heater is energized for at least 12 hours before the compressor is started. (Disconnect switch on and wall thermostat off.)

**TABLE 8**  
MAXIMUM SYSTEM CHARGE VALUES (-)ANL

| (-)ANL<br>Model<br>Size | Compressor<br>Model Number | Charge Limit<br>Without Crankcase<br>Heat (3 Phase) | Charge Limit<br>Without Crankcase<br>Heat (1 Phase) |
|-------------------------|----------------------------|-----------------------------------------------------|-----------------------------------------------------|
| 18                      | ZP16K5E                    | 8 lbs.                                              | 9.6 lbs.                                            |
| 24                      | ZP21K5E                    | 8 lbs.                                              | 9.6 lbs.                                            |
| 31                      | ZP25K5E                    | 8 lbs.                                              | 9.6 lbs.                                            |
| 37                      | ZP34K5E                    | 10 lbs.                                             | 12 lbs.                                             |
| 43                      | ZP38K5F                    | 10 lbs.                                             | 12 lbs.                                             |
| 49                      | ZP42K5E                    | 10 lbs.                                             | 12 lbs.                                             |
| 60                      | ZP54K5E                    | 10 lbs.                                             | 12 lbs.                                             |

**TABLE 9**  
MAXIMUM SYSTEM CHARGE VALUES (-)APM

| (-)APM<br>Model<br>Size | Compressor<br>Model Number | Charge Limit<br>Without Crankcase<br>Heat (1 Phase) |
|-------------------------|----------------------------|-----------------------------------------------------|
| 18                      | ZP16K5E                    | 9.6 lbs.                                            |
| 24                      | ZP20K5E                    | 9.6 lbs.                                            |
| 30                      | ZP24K5E                    | 9.6 lbs.                                            |
| 36                      | ZP31K5E                    | 9.6 lbs.                                            |
| 42                      | ZP34K5E                    | 12 lbs.                                             |
| 48                      | ZP42K5E                    | 12 lbs.                                             |
| 56                      | ZP44K5E                    | 12 lbs.                                             |
| 60                      | ZP51K5E                    | 12 lbs.                                             |

## **HARD START COMPONENTS**

Start components are not usually required with the scroll compressors used in (-)ANL/(-)APM condensing units, but are available for special cases and where start components are desirable to reduce light dimming.

## **TIME DELAY CONTROL (TDC)**

The time delay (TDC) is in the low voltage control circuit. When the compressor shuts off due to a power failure or thermostat operation, this control keeps it off at least 5 minutes which allows the system pressure to equalize, thus not damaging the compressor or blowing fuses on start-up.

## **LOW AMBIENT CONTROL (LAC)**

This component senses compressor head pressure and shuts the condenser fan off when the head pressure drops below designated levels. This allows the unit to build a sufficient head pressure at lower ambient in order to maintain system balance and obtain improved capacity. Low ambient control should be used on all equipment operated below 65°F ambient.

## **OUTDOOR UNIT COVERS**

Outdoor condensing unit covers are available if the homeowner requests a cover for their unit. With the complete model number for the unit, the correct cover can be obtained through an authorized distributor.



### **CAUTION**

**FAILURE TO REMOVE CONDENSING UNIT COVER BEFORE OPERATING  
OUTDOOR UNIT CAN CAUSE COMPONENTS TO FAIL.**

## **SERVICE**

### **SINGLE-POLE COMPRESSOR CONTACTOR (CC)**

Single-pole contactors are used on all standard single phase units up through 5 tons. Caution must be exercised when servicing as only one leg of the power supply is broken with the contactor. Two pole contactors are used on some three phase units.

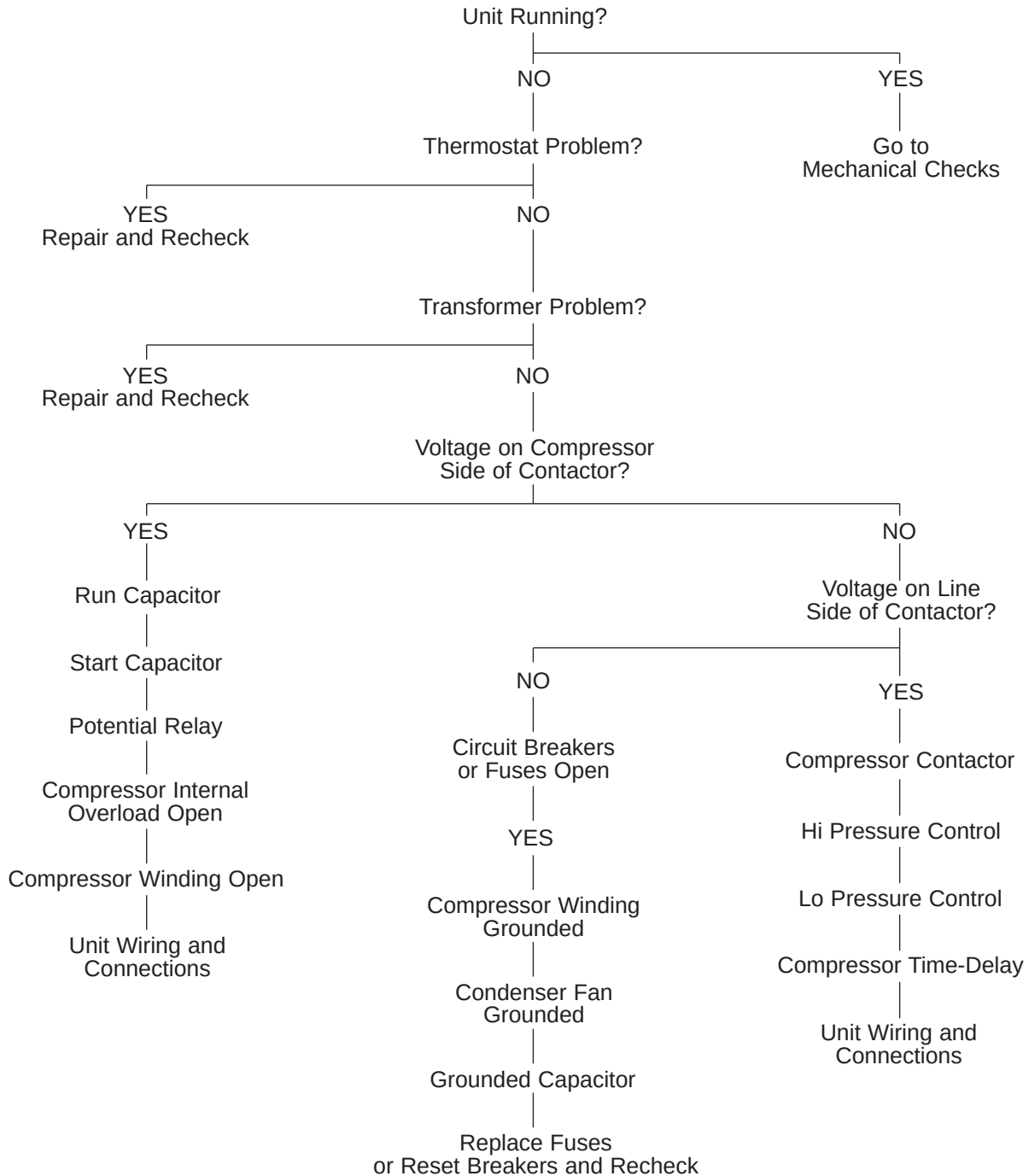


# TROUBLE SHOOTING

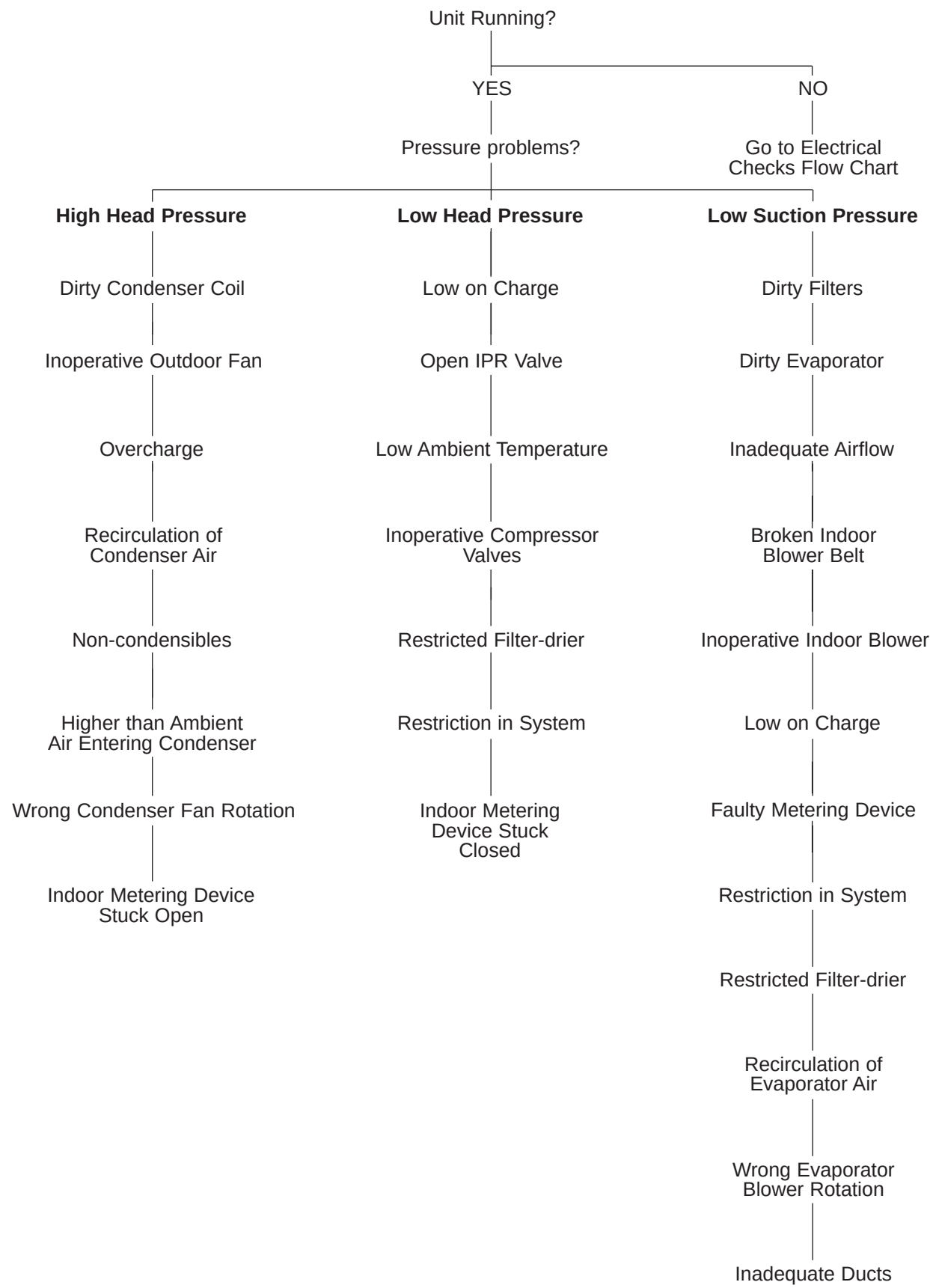
In diagnosing common faults in the air conditioning system, it is useful to present the logical pattern of thought that is used by experienced technicians. The charts which follow are not intended to be an answer to all problems, but only to guide your thinking as you attempt to decide on your course of action. Through a series of yes and no answers, you will follow the logical path to a likely conclusion.

Use these charts as you would a road map, if you are a beginning technician. As you gain experience, you will learn where to establish the shortcuts. Remember that the chart will help clarify the logical path to the problem.

## ELECTRICAL CHECKS FLOW CHART



MECHANICAL CHECKS FLOW CHART



**TABLE 10**  
TEMPERATURE PRESSURE CHART

| TEMP<br>(Deg. F) | R-410A<br>PSIG |
|------------------|----------------|
| -150             | —              |
| -140             | —              |
| -130             | —              |
| -120             | —              |
| -110             | —              |
| -100             | —              |
| -90              | —              |
| -80              | —              |
| -70              | —              |
| -60              | 0.4            |
| -50              | 5.1            |
| -40              | 10.9           |
| -35              | 14.2           |
| -30              | 17.9           |
| -25              | 22.0           |
| -20              | 26.4           |
| -15              | 31.3           |
| -10              | 36.5           |
| -5               | 42.2           |
| 0                | 48.4           |
| 5                | 55.1           |
| 10               | 62.4           |
| 15               | 70.2           |
| 20               | 78.5           |
| 25               | 87.5           |
| 30               | 97.2           |
| 35               | 107.5          |
| 40               | 118.5          |
| 45               | 130.2          |
| 50               | 142.7          |
| 55               | 156.0          |
| 60               | 170.1          |
| 65               | 185.1          |
| 70               | 201.0          |
| 75               | 217.8          |
| 80               | 235.6          |
| 85               | 254.5          |
| 90               | 274.3          |
| 95               | 295.3          |
| 100              | 317.4          |
| 105              | 340.6          |
| 110              | 365.1          |
| 115              | 390.9          |
| 120              | 418.0          |
| 125              | 446.5          |
| 130              | 476.5          |
| 135              | 508.0          |
| 140              | 541.2          |
| 145              | 576.0          |
| 150              | 612.8          |

## 23.3 SUPERHEAT CALCULATION

1. Measure the suction pressure at the suction line service valve.
2. Convert the suction pressure to saturated temperature. See Table 10.
3. Measure the temperature of the suction line at the suction line service valve.
4. Compare the temperature of the suction line to the saturated temperature.
5. The difference between saturated temperature and suction line temperature is the superheat. Superheat normal range 12° to 15°.

## 23.4 SUBCOOLING CALCULATION

1. Measure the liquid pressure at the liquid line service valve.
2. Convert the liquid line pressure to saturated temperature. See Table 10.
3. Measure the liquid line temperature at the liquid line service valve.
4. Compare the liquid line temperature to the saturated temperature.
5. The difference between saturated temperature and liquid line temperature is the subcooling. Subcooling normal range 9° to 12°.

**TABLE 11**  
AIR CONDITIONING SYSTEM TROUBLESHOOTING TIPS

| AIR CONDITIONING SYSTEM<br>TROUBLESHOOTING TIPS |                       |                     |           |            |                    |
|-------------------------------------------------|-----------------------|---------------------|-----------|------------|--------------------|
| SYSTEM PROBLEM                                  | INDICATORS            |                     |           |            |                    |
|                                                 | DISCHARGE<br>PRESSURE | SUCTION<br>PRESSURE | SUPERHEAT | SUBCOOLING | COMPRESSOR<br>AMPS |
| Overcharge                                      | High                  | High                | Low       | High       | High               |
| Undercharge                                     | Low                   | Low                 | High      | Low        | Low                |
| Liquid Restriction (Drier)                      | Low                   | Low                 | High      | High       | Low                |
| Low Evaporator Airflow                          | Low                   | Low                 | Low       | Low        | Low                |
| Dirty Condenser                                 | High                  | High                | Low       | Low        | High               |
| Low Outside Ambient Temperature                 | Low                   | Low                 | High      | High       | Low                |
| Inefficient Compressor                          | Low                   | High                | High      | High       | Low                |
| TXV Feeler Bulb Charge Lost                     | Low                   | Low                 | High      | High       | Low                |
| Poorly Insulated Sensing Bulb                   | High                  | High                | Low       | Low        | High               |

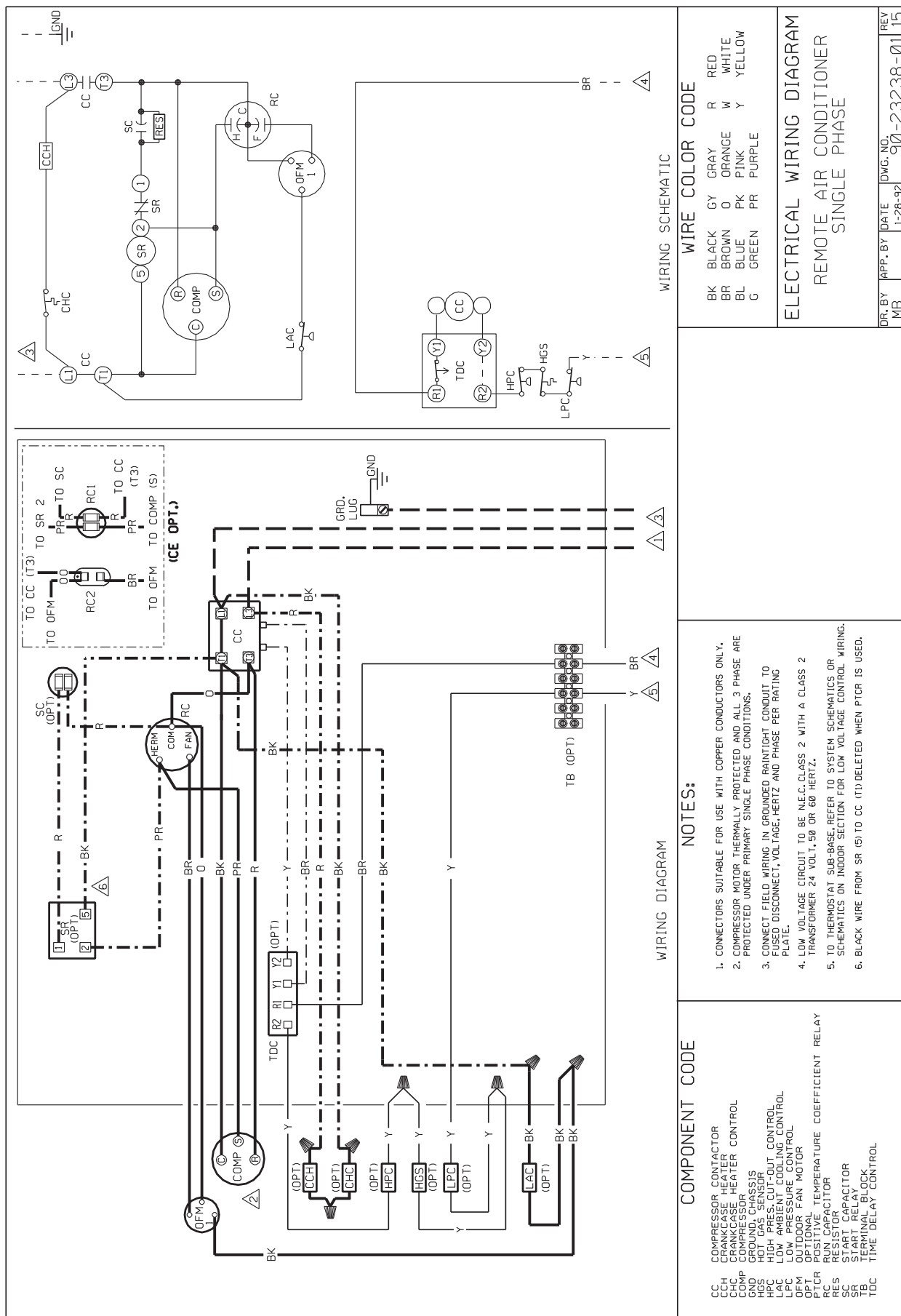
# TROUBLE SHOOTING CHART

## ▲ WARNING

**DISCONNECT ALL POWER TO UNIT BEFORE SERVICING. CONTACTOR MAY BREAK ONLY ONE SIDE. FAILURE TO SHUT OFF POWER CAN CAUSE ELECTRICAL SHOCK RESULTING IN PERSONAL INJURY OR DEATH.**

| SYMPTOM                                                    | POSSIBLE CAUSE                                                                                                                                                                                                                                                                                      | REMEDY                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit will not run                                          | <ul style="list-style-type: none"> <li>Power off or loose electrical connection</li> <li>Thermostat out of calibration-set too high</li> <li>Defective contactor</li> <li>Blown fuses / tripped breaker</li> <li>Transformer defective</li> <li>High pressure control open (if provided)</li> </ul> | <ul style="list-style-type: none"> <li>Check for correct voltage at contactor in condensing unit</li> <li>Reset</li> <li>Check for 24 volts at contactor coil - replace if contacts are open</li> <li>Replace fuses / reset breaker</li> <li>Check wiring-replace transformer</li> <li>Reset-also see high head pressure remedy-The high pressure control opens at 450 PSIG</li> </ul> |
| Outdoor fan runs, compressor doesn't                       | <ul style="list-style-type: none"> <li>Run or start capacitor defective</li> <li>Start relay defective</li> <li>Loose connection</li> <li>Compressor stuck, grounded or open motor winding, open internal overload.</li> <li>Low voltage condition</li> </ul>                                       | <ul style="list-style-type: none"> <li>Replace</li> <li>Replace</li> <li>Check for correct voltage at compressor - check &amp; tighten all connections</li> <li>Wait at least 2 hours for overload to reset. If still open, replace the compressor.</li> <li>Add start kit components</li> </ul>                                                                                       |
| Insufficient cooling                                       | <ul style="list-style-type: none"> <li>Improperly sized unit</li> <li>Improper indoor airflow</li> <li>Incorrect refrigerant charge</li> <li>Air, non-condensibles or moisture in system</li> </ul>                                                                                                 | <ul style="list-style-type: none"> <li>Recalculate load</li> <li>Check - should be approximately 400 CFM per ton.</li> <li>Charge per procedure attached to unit service panel</li> <li>Recover refrigerant, evacuate &amp; recharge, add filter drier</li> </ul>                                                                                                                      |
| Compressor short cycles                                    | <ul style="list-style-type: none"> <li>Incorrect voltage</li> <li>Defective overload protector</li> <li>Refrigerant undercharge</li> </ul>                                                                                                                                                          | <ul style="list-style-type: none"> <li>At compressor terminals, voltage must be <math>\pm 10\%</math> of nameplate marking when unit is operating.</li> <li>Replace - check for correct voltage</li> <li>Add refrigerant</li> </ul>                                                                                                                                                    |
| Registers sweat                                            | <ul style="list-style-type: none"> <li>Low indoor airflow</li> </ul>                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Increase speed of blower or reduce restriction - replace air filter</li> </ul>                                                                                                                                                                                                                                                                  |
| High head-low vapor pressures                              | <ul style="list-style-type: none"> <li>Restriction in liquid line, expansion device or filter drier</li> <li>Flowcheck piston size too small</li> <li>Incorrect capillary tubes</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>Remove or replace defective component</li> <li>Change to correct size piston</li> <li>Change coil assembly</li> </ul>                                                                                                                                                                                                                           |
| High head-high or normal vapor pressure - Cooling mode     | <ul style="list-style-type: none"> <li>Dirty outdoor coil</li> <li>Refrigerant overcharge</li> <li>Outdoor fan not running</li> <li>Air or non-condensibles in system</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>Clean coil</li> <li>Correct system charge</li> <li>Repair or replace</li> <li>Recover refrigerant, evacuate &amp; recharge</li> </ul>                                                                                                                                                                                                           |
| Low head-high vapor pressures                              | <ul style="list-style-type: none"> <li>Flowcheck piston size too large</li> <li>Defective Compressor valves</li> <li>Incorrect capillary tubes</li> </ul>                                                                                                                                           | <ul style="list-style-type: none"> <li>Change to correct size piston</li> <li>Replace compressor</li> <li>Replace coil assembly</li> </ul>                                                                                                                                                                                                                                             |
| Low vapor - cool compressor - iced indoor coil             | <ul style="list-style-type: none"> <li>Low indoor airflow</li> <li>Operating below 65°F outdoors</li> <li>Moisture in system</li> </ul>                                                                                                                                                             | <ul style="list-style-type: none"> <li>Increase speed of blower or reduce restriction - replace air filter</li> <li>Add Low Ambient Kit</li> <li>Recover refrigerant - evacuate &amp; recharge - add filter drier</li> </ul>                                                                                                                                                           |
| High vapor pressure                                        | <ul style="list-style-type: none"> <li>Excessive load</li> <li>Defective compressor</li> </ul>                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Recheck load calculation</li> <li>Replace</li> </ul>                                                                                                                                                                                                                                                                                            |
| Fluctuating head & vapor pressures                         | <ul style="list-style-type: none"> <li>TXV hunting</li> <li>Air or non-condensibles in system</li> </ul>                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Check TXV bulb clamp - check air distribution on coil - replace TXV</li> <li>Recover refrigerant, evacuate &amp; recharge</li> </ul>                                                                                                                                                                                                            |
| Gurgle or pulsing noise at expansion device or liquid line | <ul style="list-style-type: none"> <li>Air or non-condensibles in system</li> </ul>                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Recover refrigerant, evacuate &amp; recharge</li> </ul>                                                                                                                                                                                                                                                                                         |

### FIGURE 3 SINGLE-PHASE WIRING DIAGRAM



**FIGURE 4**  
**THREE-PHASE WIRING DIAGRAM (C, D & Y VOLTAGES)**

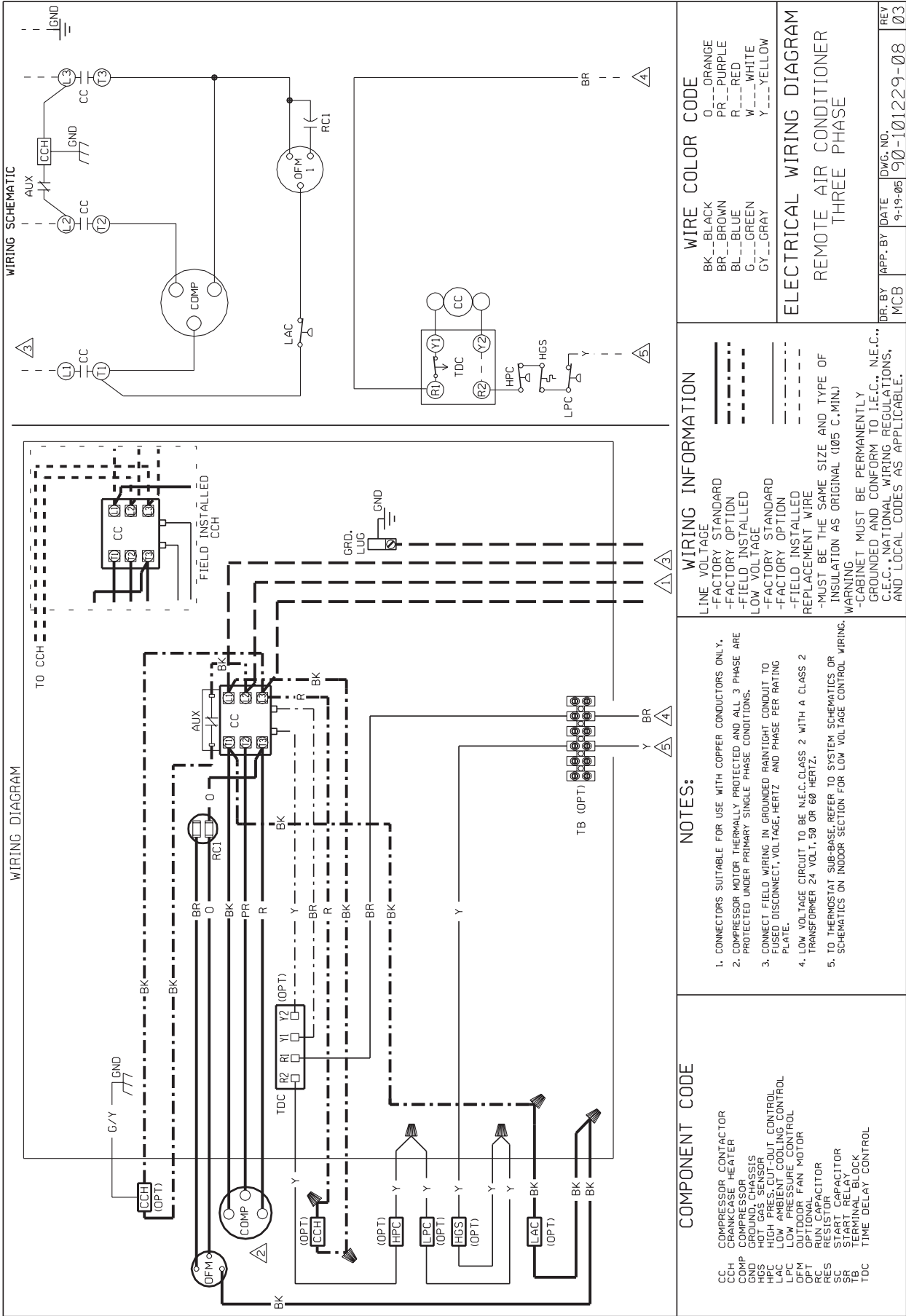
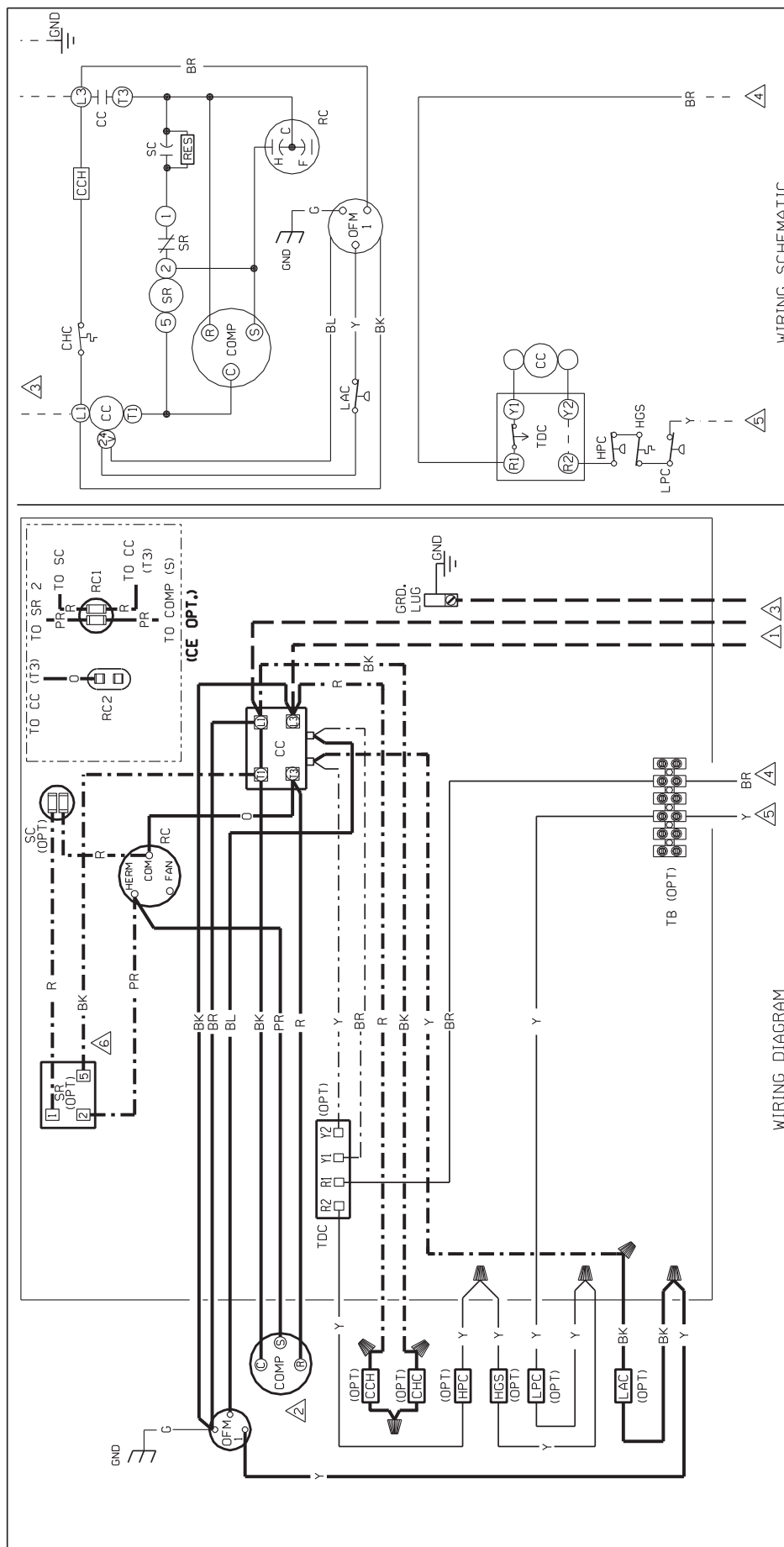


FIGURE 5



| COMPONENT CODE                                                                              | NOTES:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | WIRING INFORMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | WIRE COLOR CODE                                                                                                   |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| CC<br>CHC<br>COMP<br>GND<br>HGS<br>LAC<br>LPC<br>OFM<br>OPT<br>PC<br>RES<br>SR<br>TB<br>TDC | <ol style="list-style-type: none"> <li>CONNECTORS SUITABLE FOR USE WITH COPPER CONDUCTORS ONLY.</li> <li>COMPRESSOR MOTOR THERMALLY PROTECTED AND ALL 3 PHASE ARE PROTECTED UNDER PRIMARY SINGLE PHASE CONDITIONS.</li> <li>CONNECT FIELD WIRING IN GROUNDED RAINIGHT CONDUIT TO FIELD DISCONNECT, VOLTAGE, HERTZ AND PHASE PER RATING.</li> <li>LOW VOLTAGE CIRCUIT TO BE NEC CLASS 2 WITH A CLASS 2 TRANSFORMER 24 VOLT, 50 OR 60 HERTZ.</li> <li>TO THERMOSTAT SUB-BASE REFER TO SYSTEM SCHEMATICS OR SCHEMATICS ON INDOOR SECTION FOR LOW VOLTAGE CONTROL WIRING.</li> <li>BLACK WIRE FROM SR (5) TO CC (11) DELETED WHEN PTOR IS USED.</li> <li>IF LAC/LAR IS NOT USED, CONNECT YELLOW WIRE FROM OFM TO 24V CC.</li> </ol> | <p><b>LINE VOLTAGE</b></p> <ul style="list-style-type: none"> <li>-FACTORY STANDARD</li> <li>-FACTORY OPTION</li> <li>-FIELD INSTALLED</li> </ul> <p><b>LOW VOLTAGE</b></p> <ul style="list-style-type: none"> <li>-FACTORY STANDARD</li> <li>-FACTORY OPTION</li> <li>-FIELD INSTALLED</li> </ul> <p>REPLACEMENT WIRE</p> <p>-MUST BE THE SAME SIZE AND TYPE OF INSULATION AS ORIGINAL (105 C.MIN.)</p> <p>WARNING</p> <p>-CABINET MUST BE PERMANENTLY GROUNDED AND CONFORM TO I.E.C., N.E.C., C.E.C., NATIONAL WIRING REGULATIONS, AND LOCAL CODES AS APPLICABLE.</p> | <p><b>ELECTRICAL WIRING DIAGRAM</b></p> <p>REMOTE AIR CONDITIONER<br/>SINGLE PHASE WITH<br/>OUTDOOR ECM MOTOR</p> |

| DR. BY | APP. BY | DATE    | DWG. NO.     | REV |
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| MOR    |         | 10-7-08 | 90-103021-01 | 03  |







